Anaesthesia Associates

Booking and Billing Systems Upgrade

Request for Proposal

Final

July 2026

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Introduction

Background

Anaesthesia Associates Ltd (AA) is a Christchurch based, collectively owned, service company that provides booking and billing services for approximately 85 local anaesthetists.

Anaesthetists, working independently, have arrangements with various surgeons for both recurring and casual bookings for their services. These bookings, and the subsequent billing, are coordinated by AA who provide both mobile and web-based applications for the anaesthetists to manage their calendars and capture information about the various procedures performed.

AA have a legacy system that has been in place for many years, and they are seeking to refresh their systems to provide greater reliability and efficiencies. Many of the current processes are handled manually and are subject to bottlenecks and data quality issues.

Who should respond

AA have surveyed the market for package-based solutions and are satisfied that suitable pre-built systems to address their specific requirements are not available.

AA is anticipating that a custom-built application will be the most likely approach for their requirements. Vendors may propose a solution either entirely custom built or a mixture of pre-built and customised components based on a proprietary platform. We are flexible about the approach.

Approach

This process is intended to bring out the best solutions available, and Peritia will be available to assist vendors in understanding any aspects of the requirements.

We are approaching the process in steps:

1. Evaluating the responses received and selecting a short list of providers from the responses received to our RFP
2. Providing opportunities for the short-listed providers to interview Peritia and AA staff to discuss the requirements and opportunities
3. Requesting a presentation or demonstration workshop from short-listed suppliers
4. Selection of a preferred supplier
5. Facilitating a detailed discovery and scoping process, with the preferred supplier(s), to allow both parties to clarify the requirements, and for the preferred Service Provider(s) to develop a formal Statement of Work.

Timeline

The following table outlines an approximate timeline for the envisaged project.

Activity	Proposed Dates*
Publication of the RFP and Workshop documents	16th July 2026
Close-off date for receipt of proposals	31st July 2026
Selection of short-listed providers Notification of next steps / interviews / demonstrations	7th August 2026
Demonstration Workshops	10th - 14th August 2026
Selection of preferred partner(s)	14th August 2026
Vendor Discovery / Scoping	17th – 28th August 2026
Submission of Proposal and Statement of Work	31st August 2026
Proposal Approval	4th September 2026
Project Implementation	Sep – Oct 2026
Go-live	October 2026

* Indicative Dates - To be confirmed.

Confidentiality

This RFP, and all other material related to this process, is provided in confidence to potential Service Providers.

AA will require respondents (and all related staff) to sign formal Non-Disclosure Agreements as it may be necessary to permit limited visibility of patient, and other personally identifiable, information in the compilation of the response:

- All information is confidential and intended only for the recipients.
- The information provided will not be used for any other purpose.
- All information will be held strictly confidential and only available to those directly involved in this project.

Questions and RFP Contact

Respondents are encouraged to clarify any aspect of the RFP, or ask any questions, during the response period. We reserve the right to share any questions and answers with the other respondents.

All questions and/or contact in relation to this RFP should be directed to:

Greg Smith
Director
Peritia Limited
Email: greg.smith@peritia.co.nz
Phone: 021 477 163

Demonstration participants

We recommend that you include a Solution Architect or Senior Technical Lead in your demonstration team, to allow AA to discuss aspects of the requirements in detail.

Project Overview

High-level Requirements Summary

This is an outline of the requirements from a high-level business perspective. The requirements are covered in more detail later in this document.

The system is, at its heart, a scheduling application that then extends into timesheet capture and billing.

Schedule Management

The presentation of a calendar that allows anaesthetists to manage their availability for engagements with surgeons, typically over a rolling 4-month forward window. This calendar supports an ongoing booking iteration between surgeons' rooms, hospitals and anaesthetists (via AA). There is an industry convention that these individual booking details are referred to as Cards. The coordination of the bookings and the many changes that subsequently occur involves interactions between the three parties, over several months, right up to the day on which the procedures are scheduled.

The introduction of modern integrations is a key goal to reduce manual workloads.

Billing Management

The calculation of invoice billing is the second key functionality. Once anaesthetists have completed their data capture, and work has been checked by the office, the invoices are generated from the Card information and sent to Xero for collection. Automation is also a key goal for the design of this module.

Mobile and Web Applications

Anaesthetists primarily operate via their mobile application, with access to a similar web-based application. The mobile app displays the calendar, the Cards and procedure details. The application also provides a time-sheeting function to capture time and charge details for each Procedure and displays the availability of other anaesthetists where swaps are needed. Provision should be made to allow attachments to the schedule records.

Admin office staff also require a web application that allows them to operate on the schedule and billing management. This is primarily a set of views and permissions that allow oversight of the key processes: handling manual changes from hospitals and surgeons' rooms, and validating Card entries for billing

Reporting

There are minor reporting requirements for anaesthetists that include:

- Outstanding Balances lists, and
- Monthly activity summaries

General features / Non-Functional Requirements

A fundamental requirement is an intuitive user experience and ease of use. This is particularly important for the operational modules, where users might not be entirely comfortable with modern information systems.

Robust, flexible interfaces with other systems are vital to ensure data is entered once only and is up to date in all systems.

The solution needs to provide controls around view, access, edit rights etc. Access rights should be managed by role rather than on an individual user basis, to allow for easy management and consistent permissions for staff in like roles.

Audit trails of manual and automated actions are required.

Data Volumes

The number of invoices generated may present some challenges for Xero, however around 50% of invoices are for a small number of major contract holders (hospitals and insurers). The remainder can be thought of, in general, as one-time customers and can be archived in Xero.

Anaesthetists	~85	
Annual Invoices	28,000	p.a.

Approach to Requirements

The current systems and processes are not a reliable starting point for defining the future state.

We have taken the approach of constructing a Candidate Architecture to provide a clearer view to the key functions, processes and outputs. This is presented to save time in understanding the organisational needs and will provide an easier starting point from which to engage in the design of your solution.

Request for Proposal Response

Service Providers are requested to provide information about their services and expertise, and the products and services they would offer in support of this RFP. The aim is to clearly inform AA about the following areas:

Service Provider Information

To provide an overview of Service Provider capability in delivering this specific solution and project, along with information about the team who would likely be engaged in the delivery.

Solution Overview

Provide an overview of the solution components and architectures proposed. The Candidate Design section outlines the requirements and architectures expected to be provisioned in the solution.

Client References

Provide two Client Reference sites, describing the systems implemented.

Please provide a contact name, role, phone number and email address of a contact person at each site. These reference sites should be selected based on the relationship your organisation has developed with your client and relevance to this requirement.

Note: No contact will be made without prior agreement from you as the service provider.

Innovation / Value Add

AA is seeking innovative product and design options that will specifically address their requirements, with particular focus on the user experience and ease of use.

Approach Methodology, Solution Deployment and Timeline

Describe the proposed implementation methodology used by your organisation. This may include, but is not restricted to, your approach to design, development, deployment, project management and methodologies, change management, and training. Please provide a sufficient level of detail to allow the Professional Services estimates build-up to be understood.

AA is keen to have a new solution operational as soon as possible. Please provide an indicative timeline estimate for both the preparation and delivery based on an assumed September project start date.

Scoping/Design Phase

We are proposing to undertake a funded scoping/design phase for the preferred supplier. Please provide estimated costs and duration for this phase.

Commercial Approach and Summary of Indicative Costs

Due to the nature of the development, we recognise that there is some potential for further sales of the product in NZ. We may consider commercial arrangements in this regard, without making any commitments at this stage.

We understand that early budget estimates are problematic, however providing a range of indicative implementation costs, licence costs and any associated conditions, can be helpful for budget planning.

In your response, please provide details of the various product licencing conditions and costs, and a rate table for your standard project implementation resources (PM's, Functional Specialists etc).

The pricing schedule should include:

Licencing for modules and options. Performance constraints and risks. Tier pricing.

Professional Services: Standard rates

Estimated ongoing support costs.

Initial estimate of Professional Services, subject to review after Discovery

Draft Timeline for a Project implementation

Development methodology and expected team makeup

Pricing should include details describing how adding or removing users affects licencing, and the overall approach to managing user levels.

We will treat any Professional Services budgets as indicative, as a further opportunity to undertake a more detailed discovery will be provided for the preferred supplier.

Anaesthesia Associates

Booking and Billing Systems Upgrade

Outline of Functional Requirements

Candidate Architecture

July 2026

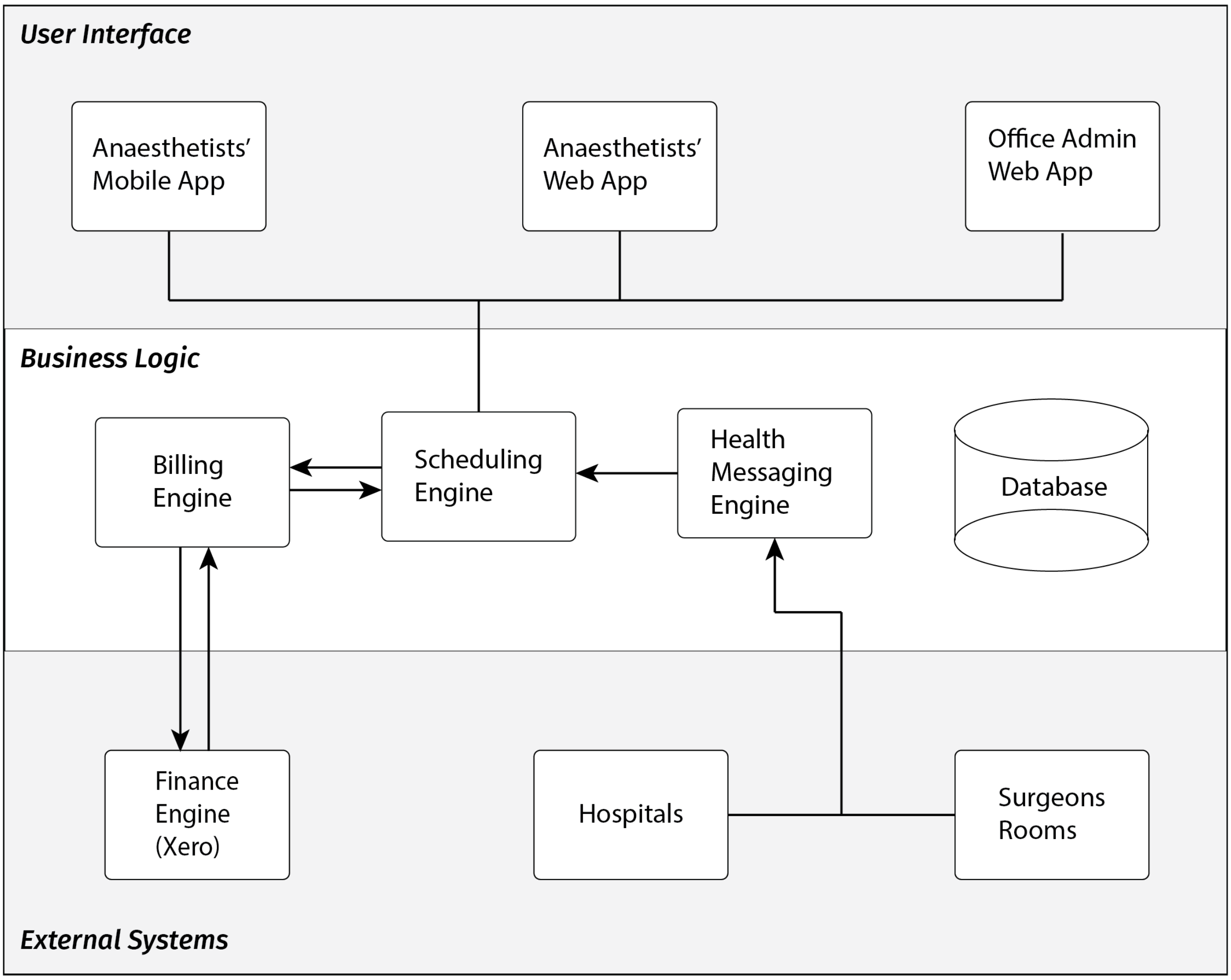
Requirements Introduction

This section introduces the AA business model and a Candidate Architecture.

AA has been using its current system for many years, and it is seeking to refresh this with an updated application that will embrace the latest technologies available to maximise the automation of their information processing, reduce staff workloads and improve data quality.

Rather than providing a Detailed Requirement Specification, we have decided to provide a technical starting point that steps into the analysis and seeks to clarify the key business functions and technical features that AA is seeking for its new systems. The conceptual system outline below is not a radical departure from the existing applications, but seeks to improve every aspect of their operation, timeliness and ease of use, and to move existing AR functions into Xero.

Major System Components



Schedule Management – Candidate Architecture

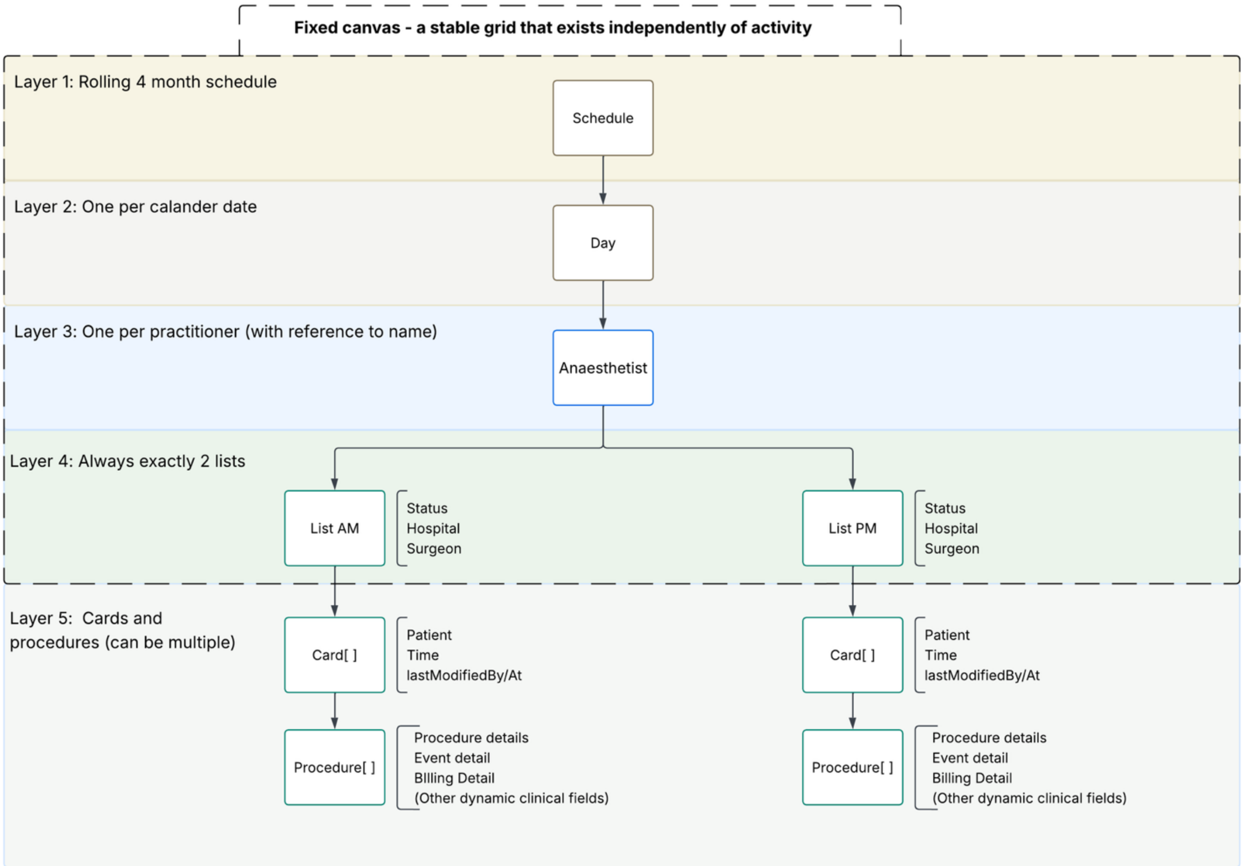
Purpose

This section describes AA's scheduling data. It is provided as a reference model — a starting point for design and discussion — not as a fixed specification. We expect this structure to be challenged, refined, and extended as the data model is developed, particularly where a different approach proves to be a better fit.

Overview

AA anaesthetists have both standing and ad-hoc arrangements with various Surgeons. These are half-day booking clusters known as Lists. Lists can be assigned automatically, based on standing arrangements (Permanent Lists), or through ad-hoc additional bookings, typically via phone from the surgeons' rooms.

The core scheduling entity is a nested hierarchy, rolling forward on a 4-month horizon:



Schedule → Day → Anaesthetist → List is structural and fixed, created by the system for every active Anaesthetist, for every Day in the rolling horizon, with exactly two Lists per Anaesthetist-Day (AM and PM). This part of the tree is effectively a canvas — a stable grid that exists independently of activity.

List Status, Hospital, Surgeon, and Cards are the data painted onto that canvas; they vary, but the canvas itself does not grow or shrink with them.

Entity Descriptions

Schedule

The top-level container for AA's forward rolling calendar. Conceptually a single ongoing structure rather than a series of discrete, separately generated calendars.

Day

A calendar date within the Schedule. Each active Anaesthetist has a presence on each Day within the rolling horizon, regardless of whether they are working.

Anaesthetist

Anaesthetists are sourced from the master Anaesthetist table and referenced into the Day structure rather than owned by it.

List

A half-day session (AM or PM) belonging to an Anaesthetist on a given Day. Every Anaesthetist has exactly two Lists on every Day within the rolling horizon — this is fixed structure, not conditional on activity. The List is a first-class object — it always exists once the schedule horizon reaches that date and carries its own status independent of whether it currently holds Cards. The start and end times of the list have a default value, but that may be overridden. All day bookings simply use both lists.

- Status reflects the current state of the List, for this Anaesthetist, — e.g. available, available for emergency, unavailable, on leave (etc) — and is sourced from a master ListStatus table. Availability is set at the List (half-day) level, not the Anaesthetist or Day level, since an Anaesthetist's availability can vary independently between AM and PM.
- Hospital is the physical location the List takes place at, drawn from a master Hospital table. Theatre information is not currently recorded.
- Surgeon is the anaesthetist's surgeon for that List. The assignment is usually (approximately 80%) defined in the anaesthetist's Permanent List but may otherwise be assigned manually by office staff. The list instance is then only available for that anaesthetist/surgeon pairing.

Card

An appointment slot within a List, filled progressively by the surgeon (via automated process, integrations or manual advice) in the weeks leading up to the procedure. A List will contain multiple Cards, displayed in time order.

- patient references the master Patient table (unique identifier: NHI number).
- time is the scheduled time of the Card within the List.
- lastModifiedBy / lastModifiedAt track the most recent change, given that Cards are mutable from multiple sources (surgeon integration, hospital integration, anaesthetist mobile app) right up to the time of the procedure.
- Card details are primarily filled by the surgeon/hospital, and (ideally) this data is passed through via a hospital integration rather than entered manually by AA staff.

- Full audit of changes is required at the Card level — every create, update, and reassignment must be logged, not just summarised via lastModifiedBy / lastModifiedAt. This reflects both the multiplicity of sources able to change a Card and the clinical/billing significance of an accurate change history.

Procedure

One or more clinical procedures may be performed within a single Card. This is where the bulk of dynamic clinical detail lives, including the data needed to support billing. A Card may have more than one Procedure, and each Procedure resolves its own billing route independently — within a single Card, one Procedure may be billed to the Hospital while another is billed to the Billable Party.

- billingRoute determines who is invoiced for this Procedure: Hospital, Billable Party, or Insurer. See Billing Route Resolution below.
- insurer (→ Insurer, nullable) is populated when billingRoute = Insurer, and may otherwise be recorded informationally when noted by the hospital under the Hospital route.
- governingContract (→ Contract) is looked up to determine the rate applied, independently of which billingRoute was resolved. See Billing Route Resolution below.
- priceOverride (optional) allows a discretionary adjustment to the standard rate — e.g. a friends-and-family discount — independent of any Contract. Should carry a reason/note, both for office reference and for the audit trail. Designs should allow for both agreed, fixed fees and $ or % adjustments.
- billableParty (→ Patient by default) identifies who is responsible for payment when billingRoute = Billable Party. In the common case this is the patient themself; an explicit override is used where the Billable Party differs from the patient — most commonly a guardian paying for a minor's uninsured procedure.
- Patient and Billable Party data is supplied by the surgeon/hospital as part of the booking. This data is captured here as the source feed for invoice generation; see the Xero Integration section for how it is consumed.

Billing Route Resolution

Each Procedure resolves to exactly one of three billing routes. This is set explicitly (by hospital advice, or by AA staff where the hospital does not specify) rather than derived from other flags:

Billing Route	Meaning
Hospital / Contract Holder	The Contract Holder (usually a hospital) accepts billing for this procedure, commonly noted with Contract and insurance details. The hospital may itself be acting as an agent or insurer-like party in the underlying arrangement. To accommodate various arrangements with the Hospital a number of Contracts are established – these are effectively billing rules. A Default Contract, which has no conditions will always be present. The default contract is empty and is billed at the normal (no contract rates).
Billable Party	The patient (or their override, e.g. a guardian) is billed directly. No Contract applies; rate is standard, subject to an optional price Override.
Insurer	The named Insurer is billed directly. Only available where that Insurer accepts direct claims from AA (currently one only). Rate is governed by the Contract held for that Insurer.

The Hospital entity plays two distinct roles in this model

1. As the physical location of the theatre (List.hospital, is always present) for an active List
2. As a billing counterparty (only when billingRoute = Hospital). Both roles reference the same Hospital master record; the model does not require a second Hospital reference to express this — it is simply a matter of which relationship is being used

Rate calculation is independent of billing route, and follows a uniform lookup once the route is resolved: find the governing Contract for the relevant counterparty (Hospital or Insurer), and apply its type.

Contract type	Behaviour
1 — Reference only	No overrides. Standard rate applies (anaesthetist's own $/Base unit, RVG-referenced). This is the mandatory default Contract (Contract.Name = "Default"), held for every Hospital and every direct-billing Insurer. This Contract type is also used for Hospitals requiring a Contract reference to assist with their internal analysis, but which does not impact billing rules/pricing
2 — Agreed rate / discount	Agreed rate or discount % off standard rates.

Contract type	Behaviour
3 — Fixed price	Fixed price, independent of standard rates.

Every Hospital and every direct-billing Insurer is mandated to hold at least a default Type 1 Contract (a simple, one-off admin step). This guarantees governing Contract always resolves to a real Contract when billing Route is Hospital or Insurer — there is no “no Contract found” branch in the rating logic. Contract is looked up by counterparty (Hospital or Insurer), not by procedure type or any other dimension.

Multiple Procedures resolving to the same billing route and counterparty are billed together on a single invoice where appropriate — for example, two uninsured Procedures for the same patient. Invoice generation is therefore a grouping operation over Procedures by resolved counterparty, not a one-to-one Procedure-to-invoice mapping. This is covered in the Xero Integration section.

Care will be required in the design of the charging of multiple procedures. Various rules apply depending on the nature of the contract.

Supporting / Master Data

The following are referenced by the structure above but maintained as separate, relatively static master tables:

Master table	Description / key fields
Hospital	ID and Name. Minimal additional data at present. (A Hospital is usually, but not always the Contract Holder. See note re Bariatric Surgery)
List Status	Defined set of statuses (e.g. private, public, pre-op, holiday, unavailable, free). Fields: description, colour. (to be confirmed)
Anaesthetist	AA’s anaesthetist. Basic contact information plus registration number (ID).
Hospital Holiday Calendar	Each hospital’s own availability calendar (closures, holidays, etc.), maintained independently per hospital.
Anaesthetist Availability	Maintained by each Anaesthetist via the mobile app. Reflects their personal availability, independent of any List assignment. Any change in these settings is immediately reflected in the Schedule
Recurring (“Permanent”) Lists	Standing arrangements defined by Hospital, Day of Week, Anaesthetist, and AM/PM. Used to populate the rolling schedule with default Lists going forward.

Master table	Description / key fields
Surgeon	External party, called on by AA.
Patient	Identified by NHI number, plus demographics. Patient and billable-party details are supplied by the surgeon/hospital as part of the Card, rather than originated by AA.
Insurer	Held as a proper reference table even though only one Insurer (NIB) currently accepts direct claims from AA. Key field: acceptsDirectClaims. Holds a mandatory default Type 1 Contract.
Contract	Governs rate calculation for a billing counterparty. Scoped to either a Hospital or an Insurer (an Insurer-scoped Contract applies across all hospitals). Every Hospital and every direct-billing Insurer is mandated to hold at least a default Type 1 (reference only) Contract; types 2 (agreed rate/discount) and 3 (fixed price) are used where a real negotiated arrangement exists. See Billing Route Resolution.

Key Design Principles

1. Schedule → Day → Anaesthetist → List is a fixed canvas. It is pre-populated on a rolling 4-month horizon: every active Anaesthetist has exactly two Lists (AM, PM) projected forward for every Day in the horizon, regardless of whether those Lists are in use. This canvas does not grow or shrink with activity — it is what allows AA's calendar view to render a complete picture, including unopened/free slots, without gaps. At current roster size this is a lightweight dataset (~85 anaesthetists × ~120 days × 2 lists ≈ 20,000 List records).
2. If an Anaesthetist is added, then their forward schedule needs to be populated so that the "canvas" is ready to accept bookings.
3. Status, Hospital, Surgeon, and Cards are painted onto the canvas, not part of its shape. A List's status is meaningful on its own (e.g. "this anaesthetist is on leave Tuesday PM") even with no Cards attached — status is not derived from Card activity, and the presence or absence of Cards never changes the canvas structure itself.
4. The List is the unit of availability, not the Anaesthetist or the Day. This is what supports AA's primary operational use case: when a surgeon's office calls looking for a free slot, staff need to see availability at AM/PM granularity across all anaesthetists for a given day.
5. Master/reference data is decoupled from the schedule tree. Hospital, Surgeon, Anaesthetist, Patient, and List Status, Master Calendar, Hospital Calendar and Anaesthetist Availability (calendar) are maintained independently and referenced by ID, not nested/owned within the schedule structure. This keeps the schedule tree itself relatively lightweight.

6. Recurring availability and standing arrangements sit outside this tree, but actively project into it. The Permanent List master (Hospital, Day of Week, Anaesthetist, AM/PM) is what populates the rolling canvas with default Lists as the 4-month horizon advances. It is the template; the Schedule tree is the generated instance.
7. Anaesthetist Availability and Hospital Holiday data are independent inputs, not structural members. Both are maintained on their own master calendars — availability by the Anaesthetist via mobile app, holidays by each Hospital (admin input) — and are expected to be reconciled against the canvas (e.g. flagged conflicts, surfaced as constraints) rather than merged into the List record itself.
8. Lists must support reassignment between Anaesthetists, including at short notice. Illness and other late changes mean a List — along with its existing Cards — may need to move from one Anaesthetist to another close to, or on, the day of the procedure. The data model needs a clean way to reassign a List's owning Anaesthetist without disturbing its Cards, status history, or audit trail.
9. Cards remain dynamic until the day of the procedure. Patient Cards within a List are not finalised once created — they can be added, amended, or reassigned right up to the procedure date, from multiple sources (surgeon integration, hospital integration, anaesthetist mobile app). This reinforces the last Modified By / last Modified At requirement at the Card level and has implications for concurrency handling (see Open Questions).
10. A full audit trail is a structural requirement, not an incidental field. Card and Procedure data — including billing route, governing Contract, and any price override — must support a complete history of changes (who, what, when), not just a single last Modified By / last Modified At summary. This is driven by both the multiplicity of sources able to change this data and its clinical/billing significance: an invoice generated today must be reproducible against what was true at the time it was raised, even if the underlying rules (e.g. a Contract's terms) change later. This points toward an append-only change log or event history for these entities, rather than simple mutable records with audit columns — a choice that should be made deliberately as part of the technical design, rather than left implicit.

Open Questions

- How does the data model handle a Card or Procedure being modified concurrently by multiple sources (surgeon integration, hospital integration, anaesthetist mobile app)?
- What is the precise mechanism for reassigning a List between Anaesthetists at short notice, and how is its Card, status, and audit history preserved through that reassignment?
- How should Anaesthetist Availability and Hospital Holiday data be reconciled against the schedule canvas — as hard constraints, soft warnings, or something else?
- Should the Insurer billing route eventually support its own rate table structure distinct from the Hospital Contract model, if direct-billing arrangements with insurers grow more complex than the current single-Insurer, Type 1 default case?

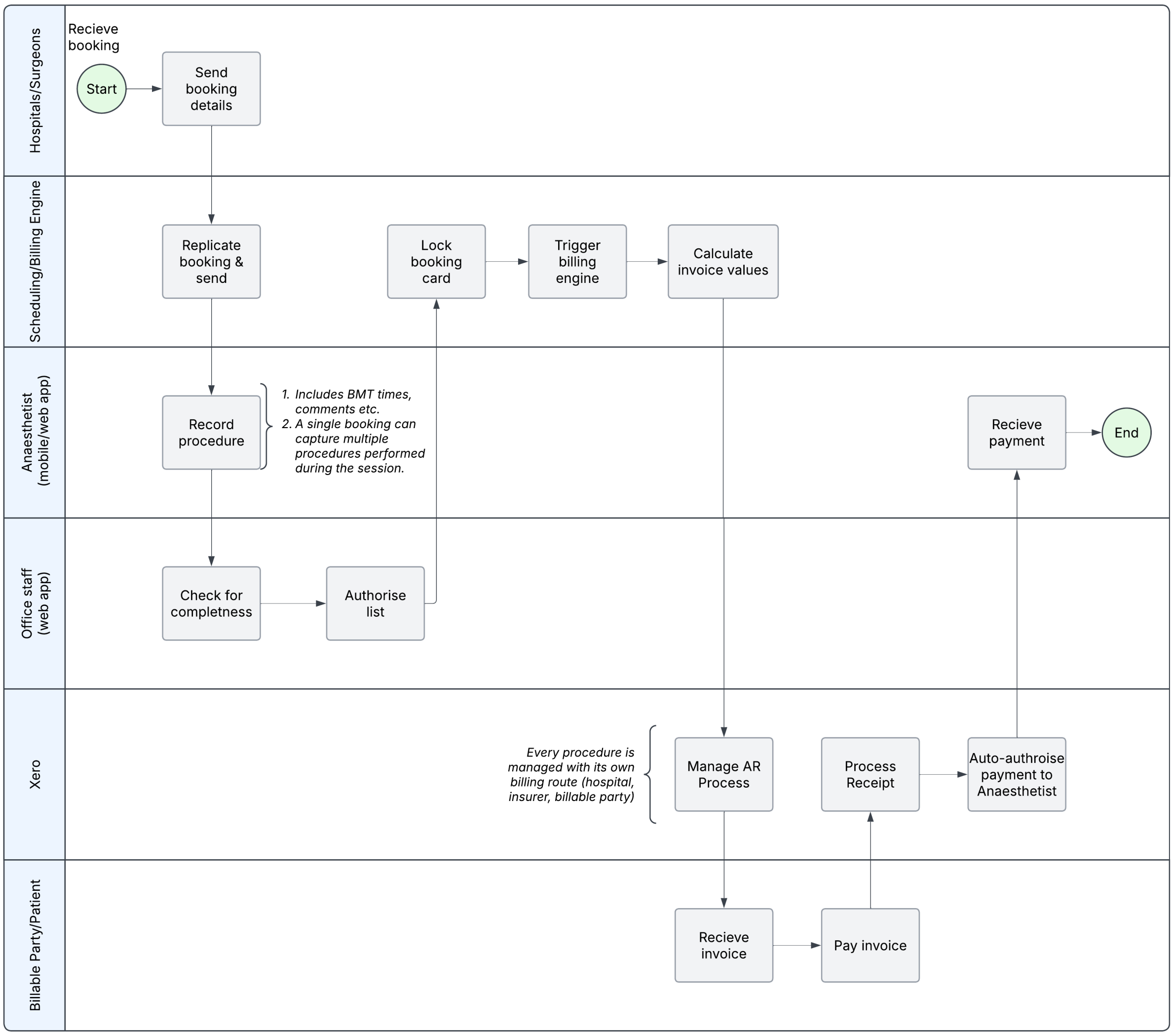
- What happens if a hospital accepts billing for a Procedure (billing Route = Hospital) but subsequently disputes or fails to pay — is there a defined fallback to the Billable Party, or is the Hospital route final once resolved?
- How will Patient and Billable Party records be deduplicated and archived at scale, given the high proportion of single-use clients? (Addressed separately in the Patient Identity Management and Archiving section.)

Billing Engine – Candidate Architecture

Generic Process Flow Overview

Before getting into the detail, the flow below illustrates where billing sits in the wider flow of a booking procedure — initiated from either a hospital update or from a surgeons' rooms, through to an anaesthetist being paid. The diagram below is intentionally simplified; each stage is expanded on in the sections that follow.

In understanding how things in this candidate design relate, it is probably more useful to think of the Billing Engine as the centre of the universe, and Xero as a Billing "add-on" – an inverse model to the one we normally think about. A separate instance of Xero will be established and its function is to provide an A/c Receivables and Banking Service to the billing engine. This instance of Xero will not be used for the general accounting work. One practical outcome of this is that invoices will be "printed" and emailed from the billing engine, rather than from Xero. This makes it easier to support the "agency" nature of the relationship between AA and the Anaesthetists. "Agent" has a very specific legal interpretation under GST legislation.



The major system components map onto the sections of this document as follows:

- Mobile / web app — where the anaesthetist captures procedure and BTM time data on the Card; referenced throughout The Big Picture section and in the *Billing Line* model described in the Candidate Data Structure section.
- Scheduling engine — the central scheduling data store (Cards, Procedures, Lists); out of scope for this section, covered in the Candidate Data Structure section.
- Admin App – where the office team check details before invoice generation
- Billing engine — the subject of this section.
- Xero — receives finalised invoice values (as ACCREC txns) and manages accounts receivable, invoicing, and collections; referenced under Payments — Where the Money Lands and Xero Integration Design, below.

The Card as the Billing Anchor

The Card is the entity from which all billing activity stems. A Card sits within the Schedule → Day → Anaesthetist → List hierarchy defined in the Candidate Data Structure section and may contain one or more Procedures and related Billable Party information. Each Procedure may in turn carry one or more Billing Lines — the RVG, fixed-fee, or individually-arranged charges described below.

Design principle: reference, don't absorb

Consistent with the design principle that master/reference data (Hospital, Surgeon, Anaesthetist, Patient) is decoupled from the schedule tree, the Billing Engine references the Card by CardID (and each Procedure by ProcedureID) rather than owning or duplicating Card data. The Card remains a Schedule system entity; the Billing Engine's own billing records point to it.

One Card, potentially multiple invoices

Billing destination (Hospital, Insurer, or Patient/Billable Party) is resolved per Procedure, not per Card. Invoice generation is therefore a grouping operation over the Procedures within a Card, by resolved counterparty:

- Where all Procedures on a Card resolve to the same counterparty (the common case), they are billed together on a single invoice.
- Where Procedures on a Card resolve to different counterparties (uncommon, but must be supported), the Card generates multiple invoices — one per distinct counterparty. The most common example of this is where a patient has requested an additional (elective, non-insured) procedure — see Split Billing, above.

This means the relationship between Card and Invoice is properly one-to-many, even though it is 1:1 in most cases. The data model and invoice-generation logic must not assume a single invoice per Card.

Card Immutability and the List Approval Process

Cards become immutable once locked — this removes any need to reconcile late clinical corrections against already-calculated billing data, and gives the Billing Engine a stable, non-drifting reference to work from. The trigger for that lock is a business process at List level, described below.

Business process

1. The anaesthetist completes all Cards within a List, then flags the List as SUBMITTED (the button label may read ‘Completed’). This sends the List to the office for checking. A List cannot be marked SUBMITTED unless all its Cards are correctly completed.

2. The office performs a sanity check on the List — typically Contract, Insurer, and reference completeness — across all Cards within it.
3. This check is managed as a review of the set of Cards within the List (an office/admin function, not a system gate).
4. When the office marks the List AUTHORISED, it is passed to the Billing Engine for processing.

Resulting state model

The List, not the Card, is the unit that carries approval state and triggers the handoff to the Billing Engine:

List state	Trigger	Card behaviour
DRAFT	Anaesthetist actively entering/editing Cards	Fully editable by the anaesthetist. Can be updated by office and integrations.
SUBMITTED	Anaesthetist flags the List as SUBMITTED (Button Label = "Completed")	Editable only by office (OfficeAdmin role).
AUTHORISED	Office signs off after sanity check	Locked — immutable; passed to Billing Engine as a unit.

Note: To align easily with Xero, we have used Xero statuses wherever possible. The one notable exception is the SUBMITTED item, above. This is to preserve continuity for users.

The Big Picture: What Are We Actually Calculating?

An anaesthetist doesn't charge a flat fee for a procedure the way, say, a plumber quotes a job. Instead, every anaesthetic generates a fee built from three separate ingredients, each calculated differently, then added together. This three-part structure — base, time and modifiers — is often shorthand-referred to as "BTM."

> **Key concept: Relative Value Units (RVUs), not dollars**
>
> The RVG does not produce a dollar figure. It produces a number of Relative Value Units (RVUs) — a unit of "value" or "work," independent of price.
> Each individual anaesthetist sets their own dollar value per unit (this is a regulatory requirement under the Commerce Act — anaesthetists can't collectively agree on pricing). The system therefore needs to store a unit-value setting per anaesthetist, not a single global price list.
> Fee = (Base units + Time units + Modifier units) × anaesthetist's $ unit value

The three components

Component	What it represents	How it's determined
Base units	The procedure itself — a fixed value reflecting the complexity/risk of the surgical site and procedure type.	Looked up from an NZSA RVG code table, organised by anatomical site (head, spine, abdomen, vascular, etc). Values range roughly from 4 units (minor/superficial) to 20–22 units

Component	What it represents	How it's determined
		(major vascular, neurosurgery). Only one base code is charged per anaesthetic.
Time units	The duration the anaesthetist had exclusive clinical responsibility for the patient.	Tiered rate: 1 unit per 15 minutes for the first two hours, then 1 unit per 10 minutes from the third hour onward. Time runs from when the anaesthetist takes over care to handover at recovery (PACU).
Modifying units	Adjustments for patient complexity, urgency, or special circumstances.	A set of separate codes added on top — e.g. pre-assessment, extremes of age, ASA physical status score, emergency status, BMI, non-standard patient positioning, awake intubation, and post-operative care events.

Base units in detail — why this can't be a fixed lookup table

The base unit table is large and irregular by nature — it reflects decades of clinical convention rather than a clean engineering taxonomy. Two details matter for system design:

- Some codes are ranges, not fixed values — for example a code might specify "6–8 units" rather than a single number, with the exact figure left to the anaesthetist's professional judgement at the time of billing.
- A loading for patient position is sometimes already built into the base code (e.g. neurosurgery and spine codes already account for prone positioning) and sometimes added separately as a modifier — the modifier rules explicitly say "if not already accounted for", meaning the system needs to know which base codes already absorb which modifiers to avoid double-charging.

The AA business rules reflect this directly: because there is such wide variability in actual procedures performed, a fully pre-populated, rules-driven base-code selector is not considered practical. Instead, the expectation is a curated dropdown of RVG codes that the anaesthetist selects from manually — the system supports selection and recording, not automatic code derivation.

Time units in detail

Time is tiered, not linear — this is a common source of calculation errors if not modelled correctly:

Tier	Rate	Applies to
T1	1 unit per 15 minutes	First 2 hours of anaesthetic time
T2	1 unit per 10 minutes	From the start of the third hour onward

Several billable events outside the main procedure also carry their own separate time-based charges — for example pain consultations, medical transport, and HDU or ward review visits that may occur days after the original procedure. This means a single patient episode can generate more than one billable "line item" over time, not just one invoice per Card.

Modifying units in detail

Modifiers layer additional units on top of base + time, depending on the clinical circumstances. They are not interchangeable — some are simple flags (a flat addition), others depend on a score or category.

ASA score entry and the Modifiers field

On the mobile app, ASA is captured in its own field, separate from the modifier total (the **M** column, for Base/Time/Modifiers). Rather than being treated as a modifier code directly, the ASA score seeds the M field with its corresponding unit value, which the anaesthetist can then adjust to add in any other applicable modifiers. As with all fields on the Card, the seeded value is overridable.

Modifier group	Example range	Nature
Pre-assessment (PA1–PA5)	1–4 units	Phone or face-to-face assessment prior to the procedure
Age extremes (A1–A2)	1–2 units	Very young or very old patients
ASA physical status (AS1–AS4)	0–4 units	Standard anaesthetic risk classification score
Emergency (ASE)	+2 units	Flat addition for emergency/unplanned procedures
BMI (OB1–OB4)	0–3 units	Patient body mass index banding
Non-standard positioning (P1)	+2 units	Conditional — only if not already included in the base code
Awake intubation (AI1)	+2 units	Specific technique flag
Post-operative care	Varies	PACU, HDU, ICU, ward reviews, nerve catheters — separately itemised, sometimes days later

ACC billing

ACC (Accident Compensation Corporation) patients are never billed directly — ACC is always billed through a hospital or another contract holder. Confirmed information from AA clarifies that ACC arrangements are themselves standard BTT (base/time/modifier) structures, identical in shape to any other RVG-calculated billing line. ACC does not require a distinct billing path or calculation model; it is simply who the contract holder is ultimately claiming from, which is invisible to the billing engine.

One narrower exception is worth confirming with AA billing: ACC pre-operative assessment has previously been described as using its own flat-fee code set (CS250, CS260, CS70), separate from the general BTM structure. This appears to be a special case limited to pre-assessment specifically, rather than a description of ACC billing generally — worth a brief confirmation, but it does not change the model. ACC pre-operative consults are usually billed separately (TBC).

ACC Contracts are set up with the Hospital or Contract Holder to cover the billing rules for ACC-related procedures. The billing rules in these ACC Contracts are independent of the Hospital but define the billing rules for the calculation of these invoices.

How Anaesthesia Associates Calculates Invoices

The RVG above is the industry reference — but AA's own rules describe three distinct ways an invoice can be produced in practice. The system needs to support all three as parallel billing methods, not just the RVG path.

Method 1 — NZSA Relative Value Guide (RVG)

This is the standard path described above: base units + time units + modifying units, with the anaesthetist selecting the closest applicable RVG code from a dropdown. As noted above, full pre-population isn't practical given procedural variability, so the system's role here is structured data capture and calculation, not automated code suggestion.

Method 2 — Pre-determined or agreed fees

Some work is billed at a fixed fee, agreed in advance, completely independent of the RVU calculation. Examples include:

- Contract pricing — negotiated rates with a hospital, insurer, or other contract holder.
- Procedure-specific agreements — for example, an agreed fixed-fee list negotiated directly with a particular plastic surgeon for their standard procedure set.

For the system, this implies a need for an “agreed fee” or “contract price list” entity that can override or bypass the RVU calculation entirely for matching Cards, keyed by some combination of contract holder, surgeon, and/or procedure type. 2nd procedure pricing usually requires additional rules.

For future proofing, and design flexibility, the hierarchy of selection for the required billing calculation algorithm, should allow for both individual contracts and organisational contracts. Currently all contracts are at the organisational level.

Method 3 — Individual billing approaches

Individual anaesthetists may use an alternative billing method entirely — for example, a simple hourly rate — rather than the RVG formula. This is permitted, but only where the arrangement has been agreed with the relevant contract holder.

> **Resolved design: no separate billing-method concept needed**
>
> This is gated by the Contract held for the relevant counterparty (see the Candidate Data Structure section) — a Billable Party, Hospital, or Insurer must hold a Contract* that explicitly permits an individually-arranged structure such as an hourly rate before one can be used.
> Once permitted, the resulting charge is captured the same way as any other non-RVG charge: as a BillingLine with chargeBasis = rate×time, attached to the relevant Procedure. It is not a structurally distinct billing method — it reuses the same BillingLine shape already defined for ancillary, non-BTM charges (see ACC billing, above, and the Candidate Data Structure section).
> Because it is the same BillingLine shape, it flows into the mobile app through the same capture path as any other BillingLine entry, rather than requiring its own UI or workflow.

** Contract, in this sense, always refers to a construct in the Billing system that defines a set of billing rules for a particular circumstance; this is not a legal contract.*

Who Receives the Invoice?

Unlike a simple patient-pays model, AA's invoices are addressed to one of two fundamentally different categories of recipient, and the category determines the entire downstream workflow: contract holders, or patients directly. As a note: these two billing types have slightly differing invoice layouts.

In this documentation, this invoice recipient is always referred to as the Billable Party.

Contract holders

These are organisations with a standing funding or contractual arrangement covering some or all of the anaesthetist's fee.

a. Hospital-based contracts

- Southern Cross Affiliated Provider (SXAP)
- Health NZ (Te Whatu Ora)

These are typically administered through specific hospitals, including:

- St George's Hospital (STG)
- Southern Cross Hospital (SX)
- Forte Health
- Christchurch Eye Surgery (CES)

b. ACC-related contracts

- Usually managed through hospitals, billed as a standard BTM structure via that Hospital's Contract — see ACC billing, above.
- Some are held externally instead — for example by orthopaedic groups such as Canterbury Orthopaedic Surgeons (COS).

c. Other specialised contracts

- Private bariatric contracts — held by the surgeon, not the hospital.
- TMJ (jaw joint) replacement contracts — held by maxillofacial surgeons.

d. Insurance Direct

- One insurer accepts claims directly. Billing is managed as for a hospital, but the account presentation workflow is currently via an upload portal.

Patients

Where there is no contract holder, the patient is billed directly. Patients fall into one of three categories, each with a different payment workflow:

Category	Description
Self-funded (post-procedure)	Patient is invoiced and pays after the procedure has taken place.
Self-funded (full or split pre-payment required)	Payment must be collected before the procedure proceeds. Any balance is then collected as normal, after the procedure.
Insured	Patient holds private health insurance and forwards the invoice to their insurer for reimbursement (the insurer is not invoiced directly by AA in this flow).

This distinction matters for the system because the contract-holder vs. patient split determines the invoice recipient and the expected payment timeline.

Split Billing

Split billing is a specific scenario the system must detect and handle correctly, since it changes how units can be claimed.

It occurs in two situations:

1. A patient undergoes multiple procedures in the same episode of care.
2. A single procedure is only partially covered by a contract or funding arrangement (i.e. part of the fee falls to a contract holder and part to the patient, or two different funders each cover a portion).

In either case, two separate invoices must be generated rather than one combined invoice.

> **Critical business rule — do not double-charge base or modifier units**
>
> When a second (additional) procedure is billed as part of split billing:
> • Only additional time units may be claimed for that second procedure.
> • Base units and modifying units cannot be claimed again — they were already charged against the primary procedure.
> This is a hard constraint the billing engine must enforce structurally, not just as a UI hint — for example, by preventing a base-unit field from being editable on any procedure flagged as "additional" within a split-billing episode.

Payments — Where the Money Lands

Under the new system, all payments will be processed by AA. Regardless of Billable Party, all funds are received into the AA account, and AA is responsible for reconciling and disbursing to anaesthetists.

> **Change from the previous arrangement**
>
> Previously, certain hospital contract holders (St George's, Forte Health, Christchurch Eye) paid anaesthetists directly into their personal bank accounts, with a remittance advice emailed to AA for manual reconciliation. That direct-payment pathway is retired under the new system — the billing engine does not need to model two separate payment destinations going forward, only the single AA account flow.

This simplifies payment-destination logic considerably (one path, not two), but it shifts new responsibility onto the system: AA is now always the intermediary, so the system needs to track and manage anaesthetist disbursement (AA → anaesthetist) as its own first-class process, rather than something that happened invisibly outside AA's books for some contract holders.

The system treats "invoice paid (into AA account)" and "disbursed to anaesthetist" as two distinct, separately trackable states for every payment. This maps directly onto the ACCREC/ACCPAY pairing described under Xero Integration Design, below: the ACCREC represents money landing in the AA account, and the ACCPAY payables run represents the subsequent disbursement to the anaesthetist.

The system operates in the same way as a Trust account would.

Summary: What This Means for the Billing Engine

Pulling the above together, here is the set of core entities and rules a technical reader should take away from this section:

Core data the system will capture per Card

- Billing method: RVG-calculated / pre-agreed fixed fee / pre-agreed rate / individually arranged.
- If RVG: selected base code (with override for range-based codes), anaesthetic start/handover time, and applicable modifier codes.
- Invoice recipient type: contract holder (which Hospital and which Contract), or Insurer Direct (NIB), or Patient Direct.
- Split-billing flag, with structural prevention of duplicate base/modifier claims on additional procedures.
- Note that for split billing, only the T element of BTM is used for the secondary procedure(s) billing.
- Other data items determined during detailed design.

Key business logic the engine must enforce

1. RVU calculation: tiered time-unit logic (T1/T2 breakpoint at 2 hours), with base units capped at one per anaesthetic.
2. Conditional modifiers: positioning loading must check whether the selected base code already includes it before adding the P1 modifier.
3. ACC: billed referencing the Hospital (or other contract holder)'s Contract — not a distinct calculation path.
4. Split billing: second/subsequent procedures in an episode can only add time units, never base or modifier units.
5. Fixed-fee and individually-arranged methods: captured as a BillingLine with the appropriate chargeBasis (governed by Contract), rather than a separate billing-method flag — see Method 3, above.
6. Additional services/fees, added to the procedure record and billed accordingly.

Card Copy

On rare occasions the anaesthetist may want to copy a Card as a way of adding an additional procedure – this copy would be of the skeleton information on the selected Card, less any specific procedure details.

Ad-Hoc Card creation

In circumstances where there are differences between the hospital's list and the mobile device (for whatever reason, and there are many), the anaesthetist needs to be able to enter a new Card from scratch.

An "add" functionality will be required to support this operation, and the data would need to flow from the Mobile back to the scheduling engine. One mechanism requested is for the anaesthetist to capture the data with a photo of the hospital's or surgeon's Card details and have these processed by the system.

Proposed Design decisions

- No system gate is required on pre-op review (staff reviewing "tomorrow" on the Admin Web App calendar is an operational practice, not a tracked state transition).
- Cards are never sent back to the anaesthetist. There is no "Returned" state. Any issue found by the office is resolved by phone, always initiated by office staff — not by the anaesthetist re-editing a completed Card.
- Once a List reaches SUBMITTED, the anaesthetist has no edit access to its Cards at all — this is an access-control rule in the mobile app, not just a data-integrity lock. Corrections from that point are an admin/office function.
- The mobile app itself performs validation to ensure minimum data required for billing is present before a Card can be marked complete, reducing (but not eliminating) the office's sanity-check burden.
- References to "mobile app" throughout this section apply equally to the anaesthetists' web app view.

Billing Engine Integration Point

The Billing Engine's integration surface with the scheduling system is deliberately narrow: it acts on a single event — a List reaching AUTHORISED — and receives the whole List as a unit. It does not need to observe individual Card-level events.

On receipt of an AUTHORISED List, the Billing Engine:

1. Iterates the Cards within the List, and their Procedures.
2. Resolves the billing counterparty for each Procedure (Hospital / Insurer / Patient-Billable Party), per the Billing Route Resolution logic defined in the Candidate Data Structure section.
3. Groups Procedures by resolved counterparty, per Card, to determine invoice boundaries (see The Card as the Billing Anchor, above).
4. Generates and despatches/queues the invoices for despatch to the Billable party.
5. Generates the matching ACCREC/ACCPAY pair(s) in Xero (see Xero Integration Design, below).
6. Reports status back for office monitoring (see Office-facing monitoring, below).

The Agency Perspective

While the invoice is addressed to the "patient", the internal accounting revolves around the Anaesthetist. The agency (AA) is concerned with the billing in terms of each anaesthetist's "ledger" position: what is owing and what is owed.

Xero Integration – Candidate Design

Invoice creation pattern

On processing of a Card (as part of an AUTHORISED List), the Billing Engine creates a matching pair of Xero invoices for each resolved counterparty, consistent with the two-state payment model — collection into the AA account, then disbursement — described under Payments, above:

- An ACCREC, raised to the paying party (Hospital, Insurer, or Patient/Billable Party) — the collection invoice; payment lands in the AA account.
- An ACCPAY, owed to the surgeon/anaesthetist — the undiscounted payable, created in DRAFT status and not yet part of any payables run. This is effectively a form of a Buyer Generated Tax Invoice.

Both records are created at the same time, from the same Billing Engine transaction, and are linked via the Billing Engine's own case record using the returned Xero GUIDs (not natively linked within Xero itself).

Reference fields — local-to-Xero matching key

Two distinct reference needs were identified, and are kept deliberately separate:

Xero field	Populated with	Used for
InvoiceNumber	Billing Engine-generated, unique invoice number	The matching key for the automated remittance/reconciliation module — this is the field hospitals' and insurers' quote back on remittance advices, and the one an automated parser should key its lookups on. Uniqueness should be enforced via the Xero organisation setting that prevents duplicate invoice numbers.
Reference	Internal CaseReference (links back to Card/Procedure)	Internal tracing and staff-facing display only. Not relied upon for automated matching, as Xero does not enforce uniqueness on this field.

This distinction matters because the remittance module handles bulk hospital payments covering multiple invoices/surgeons — it needs a reliable, uniquely-matchable field to allocate a lump payment across the correct individual invoices. InvoiceNumber is that field; Reference is not.

Payment detection and ACCPAY approval

When the ACCREC (collection invoice) is confirmed paid, the corresponding ACCPAY is set to AUTHORISED so that it is included in the next payables run:

Detection mechanism:

- Primary: Xero webhook subscription on INVOICE events. On receipt, the Billing Engine fetches the invoice, checks Status/Payments for payment, looks up the matching case by Xero GUID, and sets the linked ACCPAY to AUTHORISED.
- Partial payments by the Billable Party are passed through for payment to the anaesthetist proportionally. Any Accounts Outstanding reports show the remaining balance due. The Accounts Payable record is adjusted in a similar fashion.

- Safety net: a scheduled reconciliation poll (daily), since webhook delivery is not guaranteed. This catches any missed or failed webhook event, and aligns with the "reappears next day" behaviour described under List visibility rules, below.
- Idempotency: both paths write through the same handler, keyed by InvoiceID, so a duplicate webhook or a poll re-detecting an already-processed payment is a no-op.

Bulk remittance and bank reconciliation — out of scope here

The handling of bulk hospital payments covering multiple invoices is local to Xero's own bank reconciliation process and the Remittance add-on, and is not a Billing Engine design concern for this section. Any items the Remittance add-on cannot automatically match are simply left as unmatched items in Xero's bank reconciliation, for manual handling there. The Billing Engine's only obligation is to have populated InvoiceNumber (see Reference fields, above) reliably enough to support that downstream matching.

AA would prefer that there was a number similarity between the ACCREC and the ACCPAY records, for simplicity of human review and matching.

Mobile App Consumption Model

Billing Engine as system of record for the app

The mobile app never queries Xero directly for balance information. Xero is concerned with the Billable Parties and what they have and have not paid. The billing engine translates this into views by anaesthetist. Instead, the Billing Engine's own database is the sole source the mobile app reads from, kept in sync with Xero via the webhook/poll mechanism described under Payment detection and ACCPAY approval, above, plus its own writes at the point of invoice creation.

The two known reporting requirements for anaesthetists are:

1. An on-demand list of account balances outstanding. For a given query, a typical result set might contain around 100 records.
2. A periodic activity summary — aligned to each anaesthetist's individually-set GST period (monthly, bi-monthly, or six-monthly) — to assist in the preparation of their GST returns: a date-ranged list showing amounts received and the GST component.

Outstanding balance view — flat list, no rollup

Anaesthetists view their outstanding balance as a flat list of individual ACCPAY invoices, with no Card-level or other aggregation/rollup. If a surgeon has a query about a specific line, they contact office staff for detail — the app is not designed to answer accounting-level questions. This significantly simplifies both the mirror data structure (no rollup logic to maintain) and the UI (a list, not a summary-with-drill-down).

List visibility rules

The mobile app's List view and outstanding-balance view are two distinct surfaces, populated from different underlying states, with a defined handover between them:

List state (anaesthetist's view)	Mobile app behaviour
DRAFT — in progress	Visible and editable.

SUBMITTED — awaiting/undergoing office check	No longer editable by the anaesthetist (admin function only). Remains visible with a visual indicator ("completed, unbilled").
AUTHORISED — sent for billing	Disappears from the anaesthetist's List view. The internal "Sent for Billing" status detail is not surfaced to the anaesthetist — the List simply drops off, and the invoices would reappear the next day as line item(s) in the outstanding-balance view once the Billing Engine has generated the invoice(s). This supports the metaphor of the List as an anaesthetist's "to-do" list: items drop out of view once they are completed.

The precise trigger for a List disappearing from view should be invoice generation (i.e. submission to Xero), not the List reaching AUTHORISED and not payment — this needs to be an unambiguous system event the mobile app can key off and is called out here as a build detail to confirm rather than leave implicit.

Office-facing monitoring (distinct from the anaesthetist view)

Unlike anaesthetists, office staff need to monitor the List → Billing Engine → Xero flow for completion and errors — this is a genuine requirement, not a by-product of the other two designs, and needs its own scoped surface. Open questions to resolve during detailed design:

- Does a Card-level failure within a List (e.g. a Contract lookup that failed despite passing the office's own sanity check) block the whole List's processing, or only that Card?
- Is this monitoring surface part of the existing Admin Web App (alongside the calendar/List review function), or a separate Billing Engine administration screen?

Open Items Carried Forward

Consolidated across this section, for confirmation with AA / resolution during detailed design:

- The dropdown of RVG codes and values is curated centrally (by AA) but is not encyclopaedic, and may be overridden by the anaesthetist for a given procedure — confirm this is the intended default.
- Card-level vs List-level processing failure handling (see Office-facing monitoring, above).
- Where office monitoring of the billing flow should live — Admin Web App vs a dedicated Billing Engine admin surface (see Office-facing monitoring, above).
- Confirming the precise system event that removes a List from the anaesthetist's mobile view (invoice generation vs. a later point) (see List visibility rules, above).
- Whether the Xero organisation setting preventing duplicate invoice numbers should be a mandated configuration item for the AA Xero organisation (see Reference fields, above).

Health Systems Integration

Overview

A core requirement for the Anaesthesia Associates system upgrade is the reliable, automated exchange of appointment and patient data with hospital partner systems. Currently, this integration relies on a mixture of manual processes and HL7 v2 messaging links. The HL7 integrations in place are unreliable and frequently fail, creating administrative overhead, data quality risks, and delays in downstream workflows.

The clinical workflow this integration should support is as follows:

Manual Processing

A manual integration pathway from surgeons' rooms is required where integrations are not available. Surgeons will email pdf documents of their lists, detailing the Cards. A PDF reader capability is required to read, edit and ingest this data into the schedule. Currently, this is the main pathway for data.

Automated Processing

Once a List has been reserved for a specific surgeon at a specific hospital, any updates and additions to the Card information are forwarded to Anaesthesia Associates (AA) via a hospital integration interface (above) or manually from the surgeons' rooms. This includes new patient bookings, reschedules, modifications, and cancellations originating from both the surgeon's and the hospital's patient administration system. The hospital becomes the conduit of these changes.

The system will receive, interpret, and act on those updates automatically. Achieving reliable integration, particularly via the modern FHIR standard, would materially reduce workloads, minimise manual data checking, and improve the quality and timeliness of information available to Anaesthetists.

The following sections describe the current integration landscape, the relevant message formats and standards, the regulatory context in Aotearoa New Zealand, and the technical requirements for proposals responding to this RFP.

Current State: HL7 v2 Messaging

Most hospital patient administration systems (PAS) in New Zealand currently communicate via HL7 version 2.x — a standard that has been in clinical use since the late 1980s. While widely deployed, HL7 v2 is notoriously fragile in practice. Its pipe-delimited, flat-text message format lacks the schema enforcement of modern standards, and vendor-specific customisations (Z-segments, field reuse, non-standard code sets) mean that messages from different hospital systems often require bespoke parsing logic.

Message Structure

HL7 v2 messages use a fixed set of delimiter characters — | to separate fields, ^ to separate components, & for subcomponents, and ~ for repeating fields. The result is compact but difficult to read, debug, or validate without dedicated tooling.

Example: SIU^S14 Appointment Modification Message

The following is a representative HL7 v2.3.1 SIU^S14 (Scheduling Information Unsolicited — Appointment Modification) message as received from a hospital PAS system. This message type is used to notify our system of changes to booked surgical appointments, including the procedure, timing, patient details, and allocated anaesthetists.


```
MSH|^~\&|webPAS^HL7CISINV11.05.07^L|schlhosp|iPM|tns5qyhe|20260619153109||SIU^S14|59340703|P|2.3.1|||||AUS||en^^ISO 639-1

SCH||1661243^OAM|||||4827^TRANSFORAMINAL LUMBAR INTERBODY FUSION- TWO OR MORE LEVELS|||MIN^Minute(TI)^ISO+|||||||||

PID|1||XXX5593^^^^MR^7777||PATIENT^NAME^XXXX^^MRS^^L||19520105|F|""||9 XXXXXXX^""^""^XXXXXX 7608^2800^NEW ZEALAND^C|2800|...

AIS|1|U||20260629133000|0||240|MIN^Minute(TI)^ISO+||BOOKED

AIP|1|U|12952^SURGEON^NAME|NS^Neuro Surgery^SPECT|PROC|20260629133000|||240|MIN^Minute(TI)^ISO+||BOOKED

AIP|2|U|49641^ANAES^NAME|A^Anaesthetics^SPECT|ANAES|20260629133000|||240|MIN^Minute(TI)^ISO+||BOOKED
```


The key segments in this message are:

- MSH — Message header: source system (webPAS), destination (iPM), message type (SIU^S14), version (2.3.1), country (AUS).
- SCH — Scheduling activity: appointment ID (1661243), procedure code and description (Transforaminal Lumbar Interbody Fusion, two or more levels).
- PID — Patient identification: MRN (XXX5593), demographics, NHI-related identifiers, ethnicity (NZ European / NZHIS), address.
- AIS — Appointment slot: start time (2026-06-29 13:30), duration (240 minutes), status (BOOKED).
- AIP — Appointment personnel: surgeon (ID 12952, Neuro Surgery) and anaesthetist (ID 49641), both BOOKED.

Field-level definitions for these segments can be found in the HL7 v2.3.1 specification, freely browsable at:

https://hl7-definition.caristix.com/v2/HL7v2.3.1

A rendered HTML version of the v2.3.1 standard is also available at:

http://www.hl7.eu/HL7v2x/v231/index231.htm

Health New Zealand Policy Context: FHIR First

Health New Zealand | Te Whatu Ora, through its Health Information Standards Organisation (HISO), has established a formal FHIR-first policy that mandates the use of the HL7 FHIR (Fast Healthcare Interoperability Resources) standard in all new health data exchange solutions. The policy applies to all new builds and may also be enforced to modernise existing solutions. Implementers across the sector are required to use FHIR as the principal standard in the design of APIs and messaging services.

FHIR is identified as a key standard for interoperability alongside SNOMED CT and the International Patient Summary (IPS). From August 2025, all new integrations to national services — including the National Health Index (NHI) and Health Provider Index (HPI) — must use the FHIR API via the HNZ Digital Services Hub and enterprise Keycloak OAuth authentication.

The New Zealand FHIR Registry, co-governed by Te Whatu Ora and HL7 New Zealand, is the authoritative source of NZ FHIR profiles, implementation guides, and terminology artefacts:

https://simplifier.net/organization/nz-fhir-registry

The NZ Base FHIR Implementation Guide (currently v3.0.1, with v4 in development) defines the common extensions, profiles, identifiers, and terminologies that all NZ FHIR implementations must use as a foundation:

https://fhir.org.nz/ig/base/index.html

The FHIR Standard

What is FHIR?

FHIR (Fast Healthcare Interoperability Resources, pronounced 'fire') is HL7's current-generation interoperability standard. Unlike HL7 v2, FHIR is built on modern web architecture: resources are represented as JSON (or XML) documents and exchanged over RESTful HTTP APIs. Each clinical concept — Patient, Appointment, Anaesthetist, Observation — is defined as a discrete, schema-validated Resource with a stable URL-based identity.

FHIR R4 (Release 4) is the version mandated by Health NZ and is the basis of all current NZ implementation guides. FHIR R5 has been published but NZ has not yet migrated to it.

FHIR Schemas and Reference Documentation

The authoritative FHIR R4 specification, including all resource definitions, data types, search parameters, and operation definitions, is published at:

https://hl7.org/fhir/R4/

Key resources relevant to this integration:

- Appointment — https://hl7.org/fhir/R4/appointment.html
- Patient — https://hl7.org/fhir/R4/patient.html
- Anaesthetist — https://hl7.org/fhir/R4/practioner.html
- Bundle (transaction) — https://hl7.org/fhir/R4/bundle.html

NZ-specific profiles extending these base resources — including NHI-linked Patient identifiers and NZ ethnicity extensions — are defined in:

NZ Base Implementation Guide v3.0.1

Standards Comparison

Standard	Format	Readability	NZ Adoption	Status
HL7 v2.x	Pipe-delimited	Poor	Widespread (legacy)	Incumbent
HL7 v3	XML	Moderate	Minimal	Deprecated
CDA	XML	Moderate	Niche	Stable
FHIR R4	JSON / REST	Good	Mandated (new)	Strategic

Integration Requirements

Respondents to this RFP are required to address the following integration capabilities:

HL7 v2 Inbound Processing

The solution must be capable of receiving and parsing HL7 v2.3.1 SIU messages (S12 New, S13 Reschedule, S14 Modification, S15 Cancellation) from hospital PAS systems. Given the variability of v2 implementations across hospital systems, the solution should support configurable field mapping per hospital partner.

FHIR R4 Translation

Where hospital systems support FHIR R4, the solution must be capable of consuming FHIR Appointment, Patient, and Anaesthetist resources directly via RESTful API. For hospitals still using HL7 v2, the solution should include a translation layer that converts inbound v2 messages to FHIR-equivalent internal representations, enabling consistent downstream processing regardless of the source protocol.

FHIR-Native Architecture and Message Pre-Processing

Ideally, the integration engine should operate natively using FHIR R4, with a pre-processing layer responsible for translating inbound HL7 v2 messages into FHIR format before passing them to the core system. The intent is for the system to be FHIR-native over the long term, with the HL7 v2 pre-processor serving as a transitional bridge as hospital systems progressively adopt FHIR.

Currently, HL7 v2 messages from hospital systems are forwarded to an SFTP drive and processed in batches. This introduces latency and limits the system's ability to respond to changes in a timely manner. Real-time or near-real-time message processing — where each message is acted upon as it is received — should be the target design. This is particularly important given the user requirement to support late bookings for a List right up to the commencement of the operating session.

NZ Identity Standards

All patient records must be linked to the National Health Index (NHI) number where available. Patient identity lookups must use the NHI FHIR API via the Health NZ Digital Services Hub. Ethnicity coding must use the NZHIS Level 4 code set. Anaesthetist identity should reference the Health Provider Index (HPI) where possible.

Reliability and Monitoring

Given that current HL7 v2 integrations are unreliable and frequently fail, respondents must describe their approach to message delivery guarantees, error handling, alerting, and retry logic. The solution must provide operational visibility into integration failures and support manual intervention workflows where automated processing cannot proceed.

Reference Links

The following references are provided for respondent due diligence:

- HL7 v2.3.1 Field Definitions (Caristix): https://hl7-definition.caristix.com/v2/HL7v2.3.1
- FHIR R4 Specification: https://hl7.org/fhir/R4/
- NZ Base FHIR Implementation Guide: https://fhir.org.nz/ig/base/index.html
- NZ FHIR Registry (Simplifier): https://simplifier.net/organization/nz-fhir-registry
- Health NZ Interoperability Standards (HISO FHIR-first policy): Te Whatu Ora – Interoperability Standards
- Health NZ API Standards: https://apistandards.digital.health.nz/
- NHI FHIR Implementation Guide: https://nhi-ig.hip.digital.health.nz/

Appendix 1

Changes to the National Health Index (NHI)

Health New Zealand | Te Whatu Ora is introducing a new National Health Index (NHI) number format because the existing combination range is approaching exhaustion. Respondents must confirm that their systems and implementation roadmap account for this change as part of their proposal.

Why the change is happening

The NHI is the unique identifier assigned to every person who interacts with the New Zealand health system. The current format — three letters, three numbers, and a single numeric check digit (e.g. ZAA0067) — provides a finite pool of identifiers, and that pool is running out.

What is changing

	Current format	New format
Structure	AAANNNC (3 letters, 3 numbers, 1 numeric check digit)	AAANNAX (3 letters, 2 numbers, 1 letter, 1 alphabetic check digit)
Example	ZAA0067	ACA31FM
Check digit algorithm	Modulus 24	Modulus 23
Issuance order	Sequential	Randomised (non-sequential)

The new format extends the available range by introducing a randomised seven-character sequence and an alphabetic final character, creating over 33 million additional unique identifiers. Numbers will be issued in random, rather than sequential, order — a deliberate change to improve security and privacy, including reducing the risk of identifying multiple births from sequential NHI allocation.

Impact on patients

There is no impact on the public. Existing NHI numbers are unaffected, and patients will continue to access care exactly as they do today. The new format will only be issued to new NHI registrations — for example, newborns and people interacting with the New Zealand health system for the first time.

Impact on digital health systems

All digital health software suppliers and healthcare providers are required to have their systems and devices updated and tested to support both the current and new NHI formats by 1 July 2027. This includes validating both check-digit algorithms (modulus 24 and modulus 23), correctly parsing and storing the new alphanumeric structure, and ensuring no part of the system assumes a purely numeric or sequential NHI.

Our approach to compliance

Respondents must describe their specific approach to NHI dual-format compliance — including field validation logic, dual-format regression testing, engagement with Health NZ's published technical specifications, and completion of compliance testing ahead of the 1 July 2027 deadline. Respondents must ensure all NHI-handling components of their solution — including patient registration, record matching, interfaces to external systems, and reporting — are tested against both formats well in advance of the deadline, minimising risk to continuity of care.

Design policy: separating clinical and billing identifiers

To avoid using the NHI outside its intended medical context, the proposed solution must not use the NHI as the primary identifier within the client billing application. Instead, billing records should be keyed on Xero's native Contact ID, with the NHI attached only as a cross-reference field on the Xero contact record.

In practice, this means:

- A client's record originates in the clinical system, where they are identified by their NHI as required for care delivery.
- When a billing record is created for that client in Xero, Xero assigns its own Contact ID — this becomes the primary identifier for all billing, invoicing, and financial reporting.
- The client's NHI is stored alongside the Xero Contact ID as a custom field on the Xero contact, rather than being used as the record's key.

This gives the best of both worlds: financial workflows in Xero operate entirely on Xero's own identifier, so the NHI never functions as a billing key or appears in financial reporting structures — but staff can still look up a client in Xero directly by NHI when needed, since it is retained as a searchable reference on the contact. This keeps clinical identity and financial identity cleanly separated while preserving a fast, reliable lookup path between the two systems.

This separation reflects good privacy and data minimisation practice: it confines the NHI to systems with a genuine clinical purpose, preventing it from propagating into finance, accounting, or other non-clinical platforms where it serves no functional need and would otherwise increase the surface area for misuse or accidental disclosure.

Source: Health New Zealand | Te Whatu Ora — Upcoming changes to the NHI

Appendix 2

Patient Identity Management and Archiving

Design note — Xero billing integration with bespoke Practice Management System (PMS)

Overview

The practice handles a high volume of predominantly single-use surgical clients (≈ 28,000 invoices per year, ~99% one-time). Xero has a documented soft limit of approximately 10,000 contacts per organisation before performance degrades, so a flat "one new contact per invoice, never archived" approach is not viable. The architecture below keeps Xero performant indefinitely while preserving full patient and billing history.

Identity Layers

Three separate identifiers are used, each scoped to a single system:

Identifier	Lives In	Purpose
NHI (National Health Index)	PMS only — never sent to Xero	True clinical identity. Used at intake to validate and deduplicate patients, and to check for outstanding credit/balance issues from prior episodes.
Hidden internal unique ID	Generated and maintained by PMS; linked internally to NHI	The only patient identifier exposed to Xero — stored in Xero's ContactNumber field.
Xero ContactID	Xero-generated; cached in PMS	Xero's own system key, returned on contact creation. Cached against the hidden ID for fast future lookups.

This separation means NHI — subject to Health Information Privacy Code / HISO obligations — never leaves the PMS, while Xero still receives reliable, deduplicated billing contacts.

Workflow

Intake

PMS validates the patient against NHI — confirms identity (deduplicates against prior episodes regardless of name or address changes) and checks for outstanding credit issues.

Contact Resolution

PMS checks whether it already holds a ContactID for this patient's hidden ID.

- If yes → use the cached ContactID directly (no Xero query needed).

- If no → query Xero (GET /Contacts?where=ContactNumber==“{hidden_id}”) to check whether a contact exists; if found, cache the ContactID; if not, create a new Xero contact with ContactNumber = {hidden_id} and cache the returned ContactID.

Archived Contact Handling

If the matched contact is archived, attempt to invoice directly against it via the API. To be confirmed in sandbox testing whether this requires an unarchive step first.

Outstanding Balance Check

Before billing a new episode, call GET /Invoices?ContactIDs={id}&Statuses=AUTHORISED to surface any unpaid or overdue invoices for staff attention at check-in. Decide whether to separately distinguish “open” vs “genuinely overdue” in this filter.

Invoicing

Create and raise the invoice in Xero against the resolved ContactID.

Archiving

A scheduled job (e.g. nightly or weekly) archives Xero contacts once their invoices are fully paid and a defined inactivity window has passed (e.g. 90 days with no further activity), using the API archive flag. This is done via the cached ContactID — no search required.

Outcome

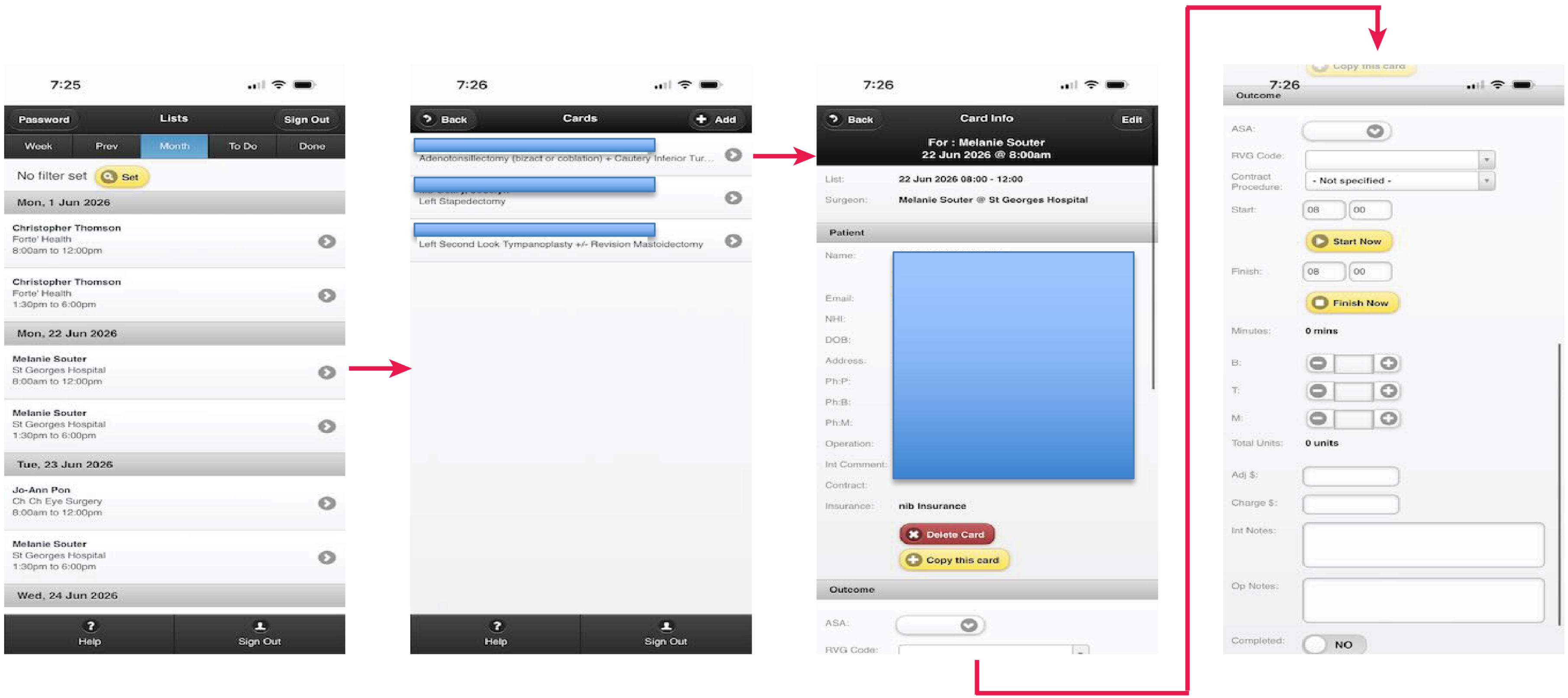
- One Xero contact per real patient, for life — true deduplication via NHI-driven matching, without name-matching fragility.
- Xero’s active contact count stays low and performant indefinitely, regardless of cumulative lifetime patient count.
- Archived contacts retain full transaction history in Xero (reports, past invoices) and can still be transacted against or restored individually if needed.
- NHI never resides in Xero, satisfying data minimisation expectations for health information.

Appendix 3

Anaesthetist Mobile App

Typical Navigation Example

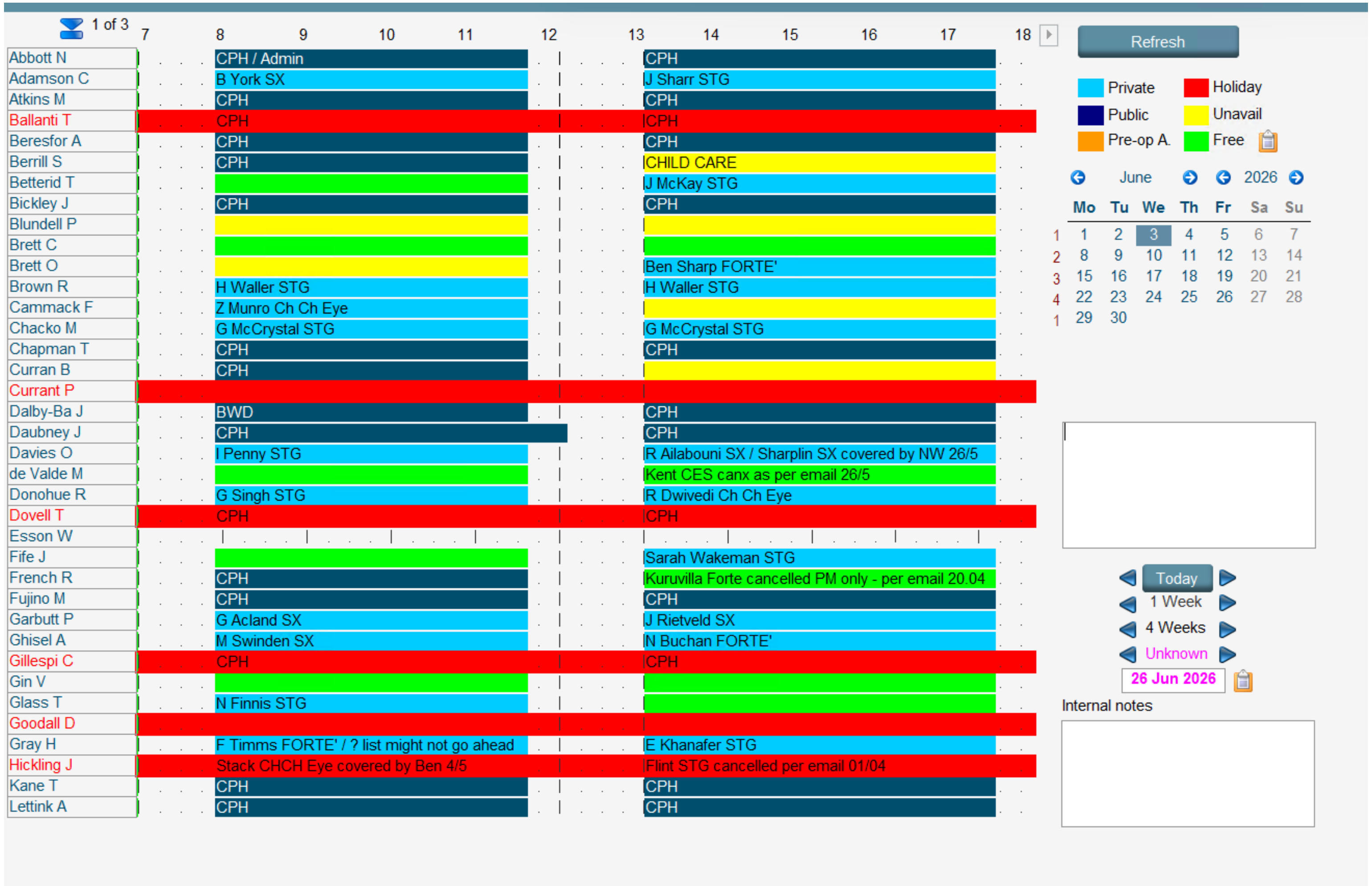
Forward Lists Cards for one list Single Card (scrolls)



Four screenshots showing the progression from schedule, to list details, to card details.

Appendix 4

Admin Schedule View (One Day Dashboard)

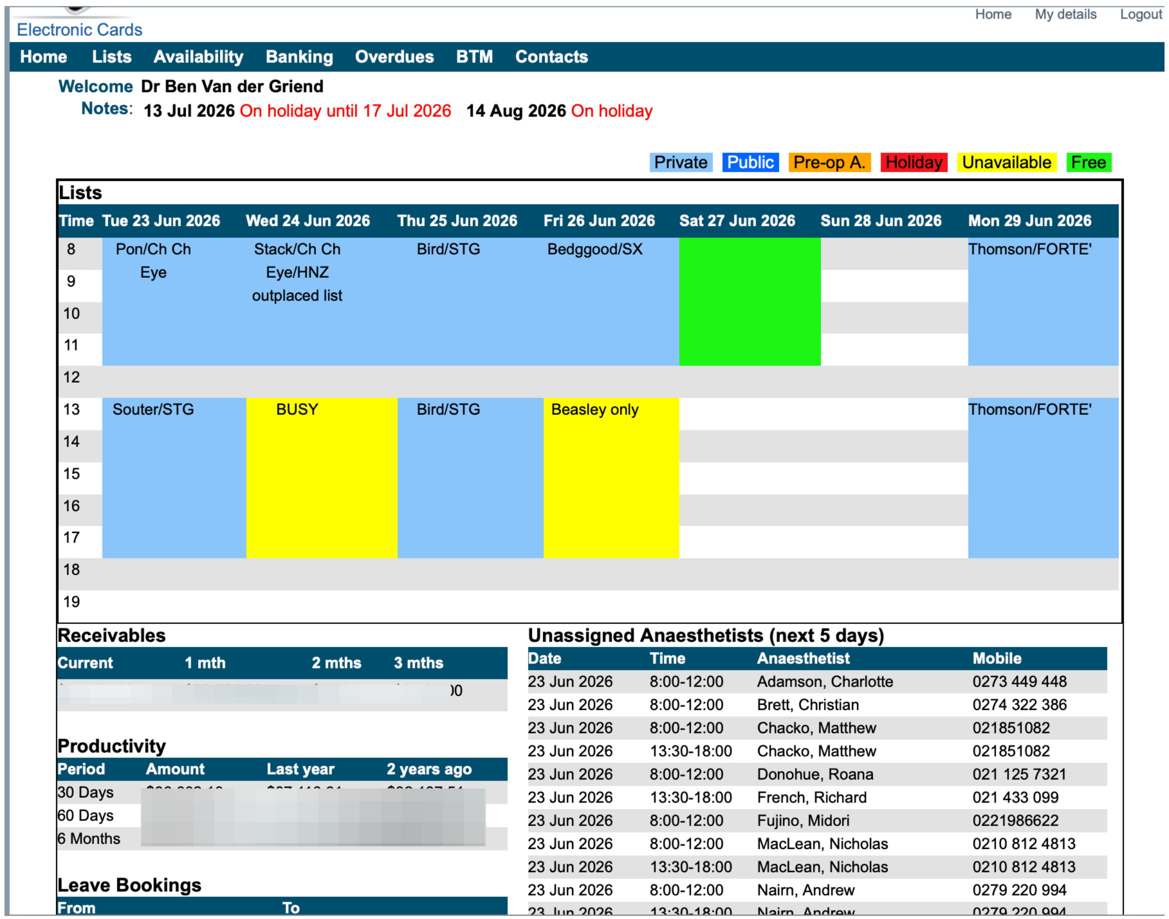


This screenshot shows the day view of the schedule. From this, users can navigate via a series of drill-downs, to detailed screens/forms.

Appendix 5

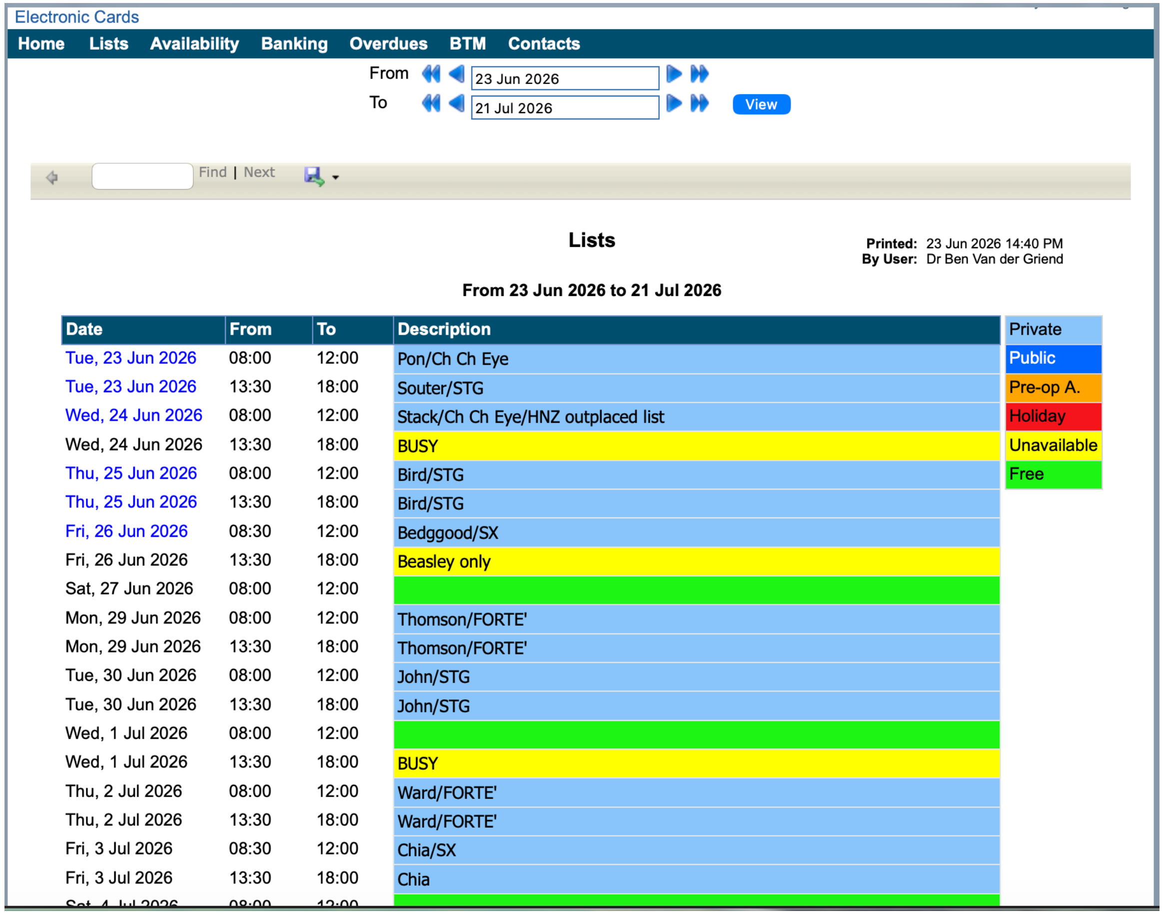
Anaesthetist Web App

Main Page – Dashboard



This is the Home page for anaesthetists, showing a summary of calendar, financial and locum availability data.

Lists


Electronic Cards

Home | Lists | Availability | Banking | Overdues | BTM | Contacts

From 23 Jun 2026
To 21 Jul 2026 View

Find | Next

Lists

Printed: 23 Jun 2026 14:40 PM
By User: Dr Ben Van der Griend

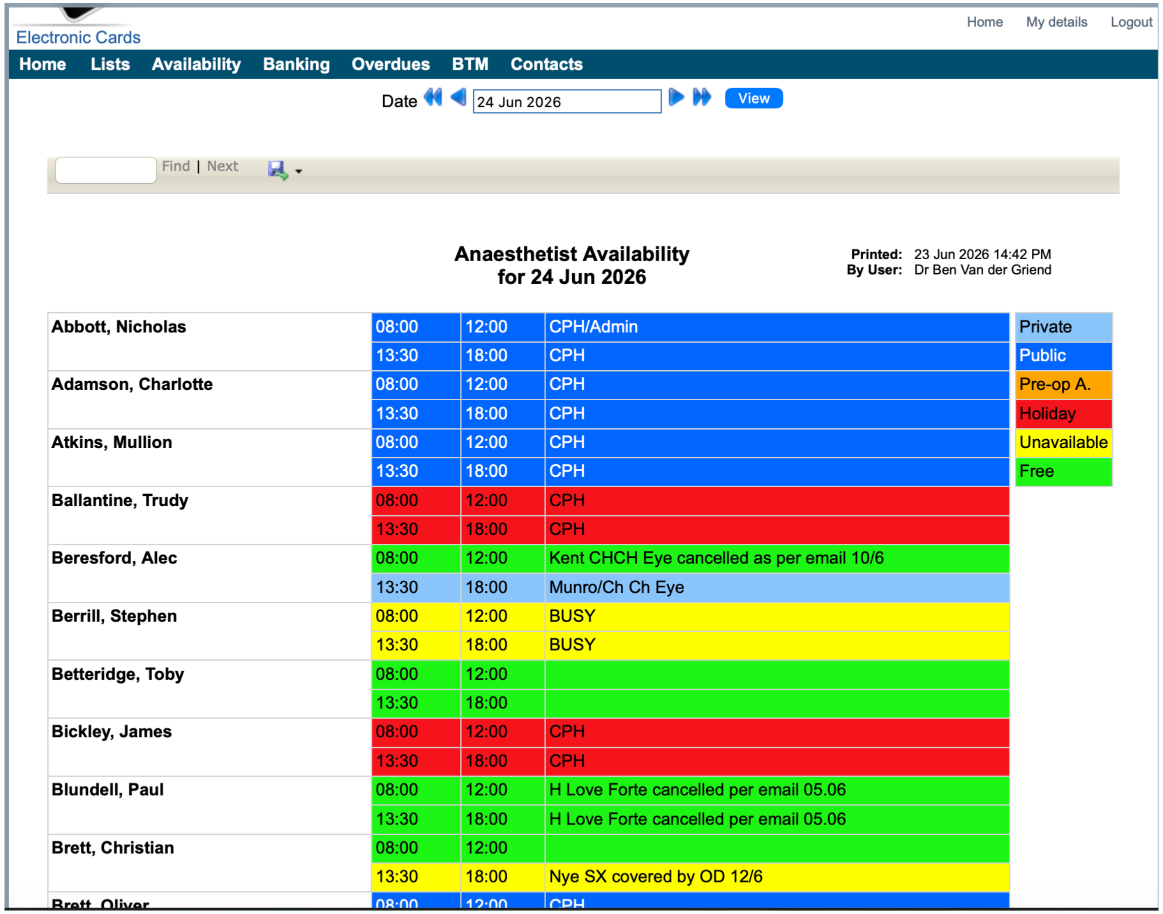
From 23 Jun 2026 to 21 Jul 2026

Date	From	To	Description
Tue, 23 Jun 2026	08:00	12:00	Pon/Ch Ch Eye
Tue, 23 Jun 2026	13:30	18:00	Souter/STG
Wed, 24 Jun 2026	08:00	12:00	Stack/Ch Ch Eye/HNZ outplaced list
Wed, 24 Jun 2026	13:30	18:00	BUSY
Thu, 25 Jun 2026	08:00	12:00	Bird/STG
Thu, 25 Jun 2026	13:30	18:00	Bird/STG
Fri, 26 Jun 2026	08:30	12:00	Bedggood/SX
Fri, 26 Jun 2026	13:30	18:00	Beasley only
Sat, 27 Jun 2026	08:00	12:00	
Mon, 29 Jun 2026	08:00	12:00	Thomson/FORTE'
Mon, 29 Jun 2026	13:30	18:00	Thomson/FORTE'
Tue, 30 Jun 2026	08:00	12:00	John/STG
Tue, 30 Jun 2026	13:30	18:00	John/STG
Wed, 1 Jul 2026	08:00	12:00	
Wed, 1 Jul 2026	13:30	18:00	BUSY
Thu, 2 Jul 2026	08:00	12:00	Ward/FORTE'
Thu, 2 Jul 2026	13:30	18:00	Ward/FORTE'
Fri, 3 Jul 2026	08:30	12:00	Chia/SX
Fri, 3 Jul 2026	13:30	18:00	Chia
Sat, 4 Jul 2026	08:00	12:00	

Private
Public
Pre-op A.
Holiday
Unavailable
Free

This page shows a simple view of upcoming lists, note the 2 lists per day repeating structure. Drill down is available.

Availability

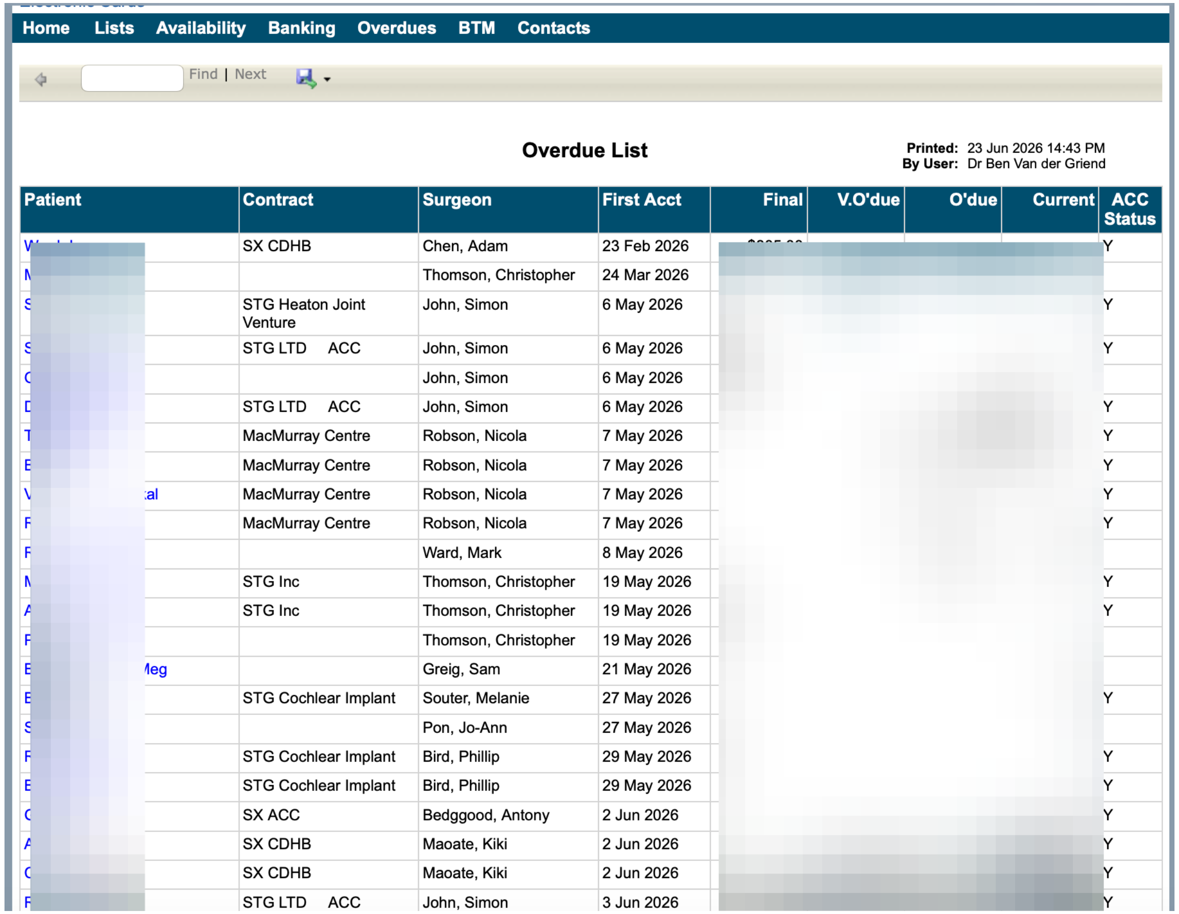


Abbott, Nicholas	08:00	12:00	CPH/Admin
	13:30	18:00	CPH
Adamson, Charlotte	08:00	12:00	CPH
	13:30	18:00	CPH
Atkins, Mullion	08:00	12:00	CPH
	13:30	18:00	CPH
Ballantine, Trudy	08:00	12:00	CPH
	13:30	18:00	CPH
Beresford, Alec	08:00	12:00	Kent CHCH Eye cancelled as per email 10/6
	13:30	18:00	Munro/Ch Ch Eye
Berrill, Stephen	08:00	12:00	BUSY
	13:30	18:00	BUSY
Betteridge, Toby	08:00	12:00	
	13:30	18:00	
Bickley, James	08:00	12:00	CPH
	13:30	18:00	CPH
Blundell, Paul	08:00	12:00	H Love Forte cancelled per email 05.06
	13:30	18:00	H Love Forte cancelled per email 05.06
Brett, Christian	08:00	12:00	
	13:30	18:00	Nye SX covered by OD 12/6
Brett, Oliver	08:00	12:00	CPH

This screen is typically used to find a replacement (locum) anaesthetist at short notice (e.g. cover for illness)

Overdue

A view of unpaid accounts



Overdue List

Printed: 23 Jun 2026 14:43 PM
By User: Dr Ben Van der Griend

Patient	Contract	Surgeon	First Acct	Final	V.O'due	O'due	Current	ACC Status
[illegible]	SX CDHB	Chen, Adam	23 Feb 2026					Y
[illegible]		Thomson, Christopher	24 Mar 2026					
[illegible]	STG Heaton Joint Venture	John, Simon	6 May 2026					Y
[illegible]	STG LTD ACC	John, Simon	6 May 2026					Y
[illegible]		John, Simon	6 May 2026					
[illegible]	STG LTD ACC	John, Simon	6 May 2026					Y
[illegible]	MacMurray Centre	Robson, Nicola	7 May 2026					Y
[illegible]	MacMurray Centre	Robson, Nicola	7 May 2026					Y
[illegible]	MacMurray Centre	Robson, Nicola	7 May 2026					Y
[illegible]	MacMurray Centre	Robson, Nicola	7 May 2026					Y
[illegible]		Ward, Mark	8 May 2026					
[illegible]	STG Inc	Thomson, Christopher	19 May 2026					Y
[illegible]	STG Inc	Thomson, Christopher	19 May 2026					Y
[illegible]		Thomson, Christopher	19 May 2026					
[illegible]		Greig, Sam	21 May 2026					
[illegible]	STG Cochlear Implant	Souter, Melanie	27 May 2026					Y
[illegible]		Pon, Jo-Ann	27 May 2026					
[illegible]	STG Cochlear Implant	Bird, Phillip	29 May 2026					Y
[illegible]	STG Cochlear Implant	Bird, Phillip	29 May 2026					Y
[illegible]	SX ACC	Beddgood, Antony	2 Jun 2026					Y
[illegible]	SX CDHB	Maoate, Kiki	2 Jun 2026					Y
[illegible]	SX CDHB	Maoate, Kiki	2 Jun 2026					Y
[illegible]	STG LTD ACC	John, Simon	3 Jun 2026					Y

This screen shows a classic accounts outstanding view, ordered by date.